

Project 8

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```
library(tidyverse)
```

```
## Warning: package 'purrr' was built under R version 4.5.2
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.1      v stringr   1.5.2
## v ggplot2    4.0.0      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.2.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(gt)
```

```
## Warning: package 'gt' was built under R version 4.5.2
```

```
library(gtsummary)
```

```
## Warning: package 'gtsummary' was built under R version 4.5.2
```

```
library(haven)
```

```
library(stringr)
```

```
library(flextable)
```

```
## Warning: package 'flextable' was built under R version 4.5.2
```

```
##
```

```
## Attaching package: 'flextable'
```

```
##
```

```
## The following object is masked from 'package:gtsummary':
```

```
##
```

```
##     continuous_summary
```

```
##
```

```
## The following object is masked from 'package:purrr':
```

```
##
```

```
##     compose
```

```
library(officer)
```

```
## Warning: package 'officer' was built under R version 4.5.2
```

```
library(summarytools)
```

```
## Warning: package 'summarytools' was built under R version 4.5.2
```

```
##  
## Attaching package: 'summarytools'  
##  
## The following object is masked from 'package:tibble':  
##  
## view
```

```
library(openxlsx)
```

```
## Warning: package 'openxlsx' was built under R version 4.5.2
```

```
library(forcats)
```

```
library(dplyr)  
library(survival)  
library(survminer)
```

```
## Warning: package 'survminer' was built under R version 4.5.2
```

```
## Loading required package: ggpubr
```

```
## Warning: package 'ggpubr' was built under R version 4.5.2
```

```
##  
## Attaching package: 'ggpubr'  
##  
## The following objects are masked from 'package:flextable':  
##  
## border, font, rotate  
##  
##  
## Attaching package: 'survminer'  
##  
## The following object is masked from 'package:survival':  
##  
## myeloma
```

```
library(labelled)
```

```
## Warning: package 'labelled' was built under R version 4.5.2
```

```
library(mice)
```

```
##  
## Attaching package: 'mice'  
##  
## The following object is masked from 'package:stats':  
##  
##     filter  
##  
## The following objects are masked from 'package:base':  
##  
##     cbind, rbind
```

```
library(jomo)
```

```
select <- dplyr::select
```

Data

```
data <- readRDS("Data/data_with_score.RDS")  
data_mod <- readRDS("Data/model_data.RDS")
```

Demographic

```
data %>%  
  select(age, Sex:dgraduation) %>%  
  tbl_summary(  
    statistic = list(  
      all_continuous() ~ "{mean} ± {sd}",  
      all_categorical() ~ "{n} ({p}%)",  
    ),  
    digits = all_continuous() ~ 2,  
    missing = "no"  
  ) %>%  
  bold_labels() %>%  
  as_flex_table() -> demo_tab  
demo_tab
```

```
## Warning: fonts used in 'flextable' are ignored because the 'pdflatex' engine is  
## used and not 'xelatex' or 'lualatex'. You can avoid this warning by using the  
## 'set_flextable_defaults(fonts_ignore=TRUE)' command or use a compatible engine  
## by defining 'latex_engine: xelatex' in the YAML header of the R Markdown  
## document.
```

Characteristic	N = 414 ¹
Age of the patient	
<=40	268 (65%)
41-50	104 (25%)
51-60	35 (8.5%)
60+	7 (1.7%)
Sex of the patient	
Male	264 (64%)
Female	150 (36%)
Family H/O cancer	
	68 (16%)
Name of cancer	
alc	1 (1.6%)
AML	1 (1.6%)
bone cat	2 (3.1%)
breast	10 (16%)
Cervix	7 (11%)
CML	1 (1.6%)
Colon	8 (13%)
GL	1 (1.6%)
HCC	2 (3.1%)
Head and Neck	1 (1.6%)
Hepatobi	1 (1.6%)
Leukemia	2 (3.1%)
lever	1 (1.6%)
Lung	10 (16%)
Ovarian	4 (6.3%)
prostate	5 (7.8%)
RCC	3 (4.7%)
stomash	1 (1.6%)
thyroid	1 (1.6%)
tongue	1 (1.6%)
trocheal	1 (1.6%)
Age of diagnosis of cancer	
<45	10 (16%)
45-55	35 (55%)
>55	19 (30%)

Characteristic	N = 414 ¹
Department	
Non Clinical	50 (16%)
Clinical	256 (84%)
Post	
Professor	18 (4.3%)
Associate Professor	41 (9.9%)
Assistant Professor	58 (14%)
Consulttant	28 (6.8%)
RP/RS	11 (2.7%)
Registrar	55 (13%)
AR	41 (9.9%)
MO	162 (39%)
Duration of graduation (years)	
<6	90 (29%)
6-10	95 (30%)
>10	127 (41%)
¹ n (%)	

Knowledge Table

```
data %>%
  select(dx_self_cancer:att_ca_scrn) %>%
  tbl_summary(
    statistic = list(
      all_continuous() ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%)",
    ),
    digits = all_continuous() ~ 2,
    missing = "no"
  ) %>%
  bold_labels() %>%
  as_flex_table() -> know_tab
know_tab
```

```
## Warning: fonts used in 'flextable' are ignored because the 'pdflatex' engine is
## used and not 'xelatex' or 'lualatex'. You can avoid this warning by using the
## 'set_flextable_defaults(fonts_ignore=TRUE)' command or use a compatible engine
## by defining 'latex_engine: xelatex' in the YAML header of the R Markdown
## document.
```

Characteristic	N = 414 ¹
Ever been diagnosed any cancer	33 (8.0%)
Name of self cancer	
Breast	8 (24%)
Cervix	4 (12%)
CML	1 (3.0%)
colon	3 (9.1%)
HCC	1 (3.0%)
Larynx	1 (3.0%)
liver	1 (3.0%)
Lung	4 (12%)
Ovarian	4 (12%)
prostate	5 (15%)
urinary	1 (3.0%)
Duration of diagnosis of self cancer	
<5	9 (27%)
5-10	21 (64%)
>10	3 (9.1%)
Know about cancer screening	347 (84%)
Name of cancer that can be screened	
cerv,ov	1 (0.8%)
B,Col,P	1 (0.8%)
Bone	2 (1.7%)
Breast	55 (46%)
brs,crvl	1 (0.8%)
brst,cer	1 (0.8%)
brst,col	6 (5.0%)
brst,pro	1 (0.8%)
bs,cl,cr	1 (0.8%)
cer,bres	2 (1.7%)
cer,pros	1 (0.8%)
Cervix	16 (13%)
col,brst	1 (0.8%)
col,brt	1 (0.8%)
col,carv	1 (0.8%)
col,pros	2 (1.7%)

Characteristic	N = 414 ¹
colon	14 (12%)
colon,br	1 (0.8%)
colon,pr	1 (0.8%)
crv,ovr	1 (0.8%)
crvl,brs	1 (0.8%)
lung	1 (0.8%)
memogr	1 (0.8%)
pro,brst	1 (0.8%)
pros,brs	1 (0.8%)
Prostate	4 (3.3%)
RCC	1 (0.8%)
Think cancer screening will usefull	374 (90%)

¹n (%)

Socre Tables

Breast

```
data_mod %>%
  select(know_breast:prac_breast) %>%
  tbl_summary(
    statistic = list(
      all_continuous() ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p})%"
    ),
    digits = all_continuous() ~ 2,
    missing = "no",
  ) %>%
  bold_labels() -> score_breast
```

Cervical

```
data_mod %>%
  select(know_cervical:prac_cervical) %>%
  tbl_summary(
    statistic = list(
      all_continuous() ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p})%"
    ),
    digits = all_continuous() ~ 2,
    missing = "no",
  )
```

```
) %>%
bold_labels() -> score_cervical
```

```
(tbl_merge(
  tbls = list("Breast Cancer" = score_breast, "Cervical Cancer" = score_cervical),
  tab_spanner = c("**Breast Cancer**", "**Cervical Cancer**")
) %>% as_flex_table -> score_br_cer)
```

Warning: fonts used in 'flextable' are ignored because the 'pdflatex' engine is used and not 'xelatex' or 'lualatex'. You can avoid this warning by using the 'set_flextable_defaults(fonts_ignore=TRUE)' command or use a compatible engine by defining 'latex_engine: xelatex' in the YAML header of the R Markdown document.

	Breast Cancer	Cervical Cancer
Characteristic	N = 414 ¹	N = 414 ¹
Knowledge		
Poor	114 (28%)	
Good	300 (72%)	
Practice		
Poor	353 (85%)	
Good	61 (15%)	
Knowledge		
Poor		120 (29%)
Good		294 (71%)
Practice		
Poor		347 (84%)
Good		67 (16%)

¹n (%)

Prostate

```
data_mod %>%
select(know_prostate:prac_prostate) %>%
tbl_summary(
  statistic = list(
    all_continuous() ~ "{mean} ± {sd}",
    all_categorical() ~ "{n} ({p}%)"
  ),
  digits = all_continuous() ~ 2,
  missing = "no"
) %>%
bold_labels() -> score_prostate
```


Colon

```
data_mod %>%
  select(know_colon:prac_colon) %>%
  tbl_summary(
    statistic = list(
      all_continuous() ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%"
    ),
    digits = all_continuous() ~ 2,
    missing = "no"
  ) %>%
  bold_labels() -> score_colon
```

```
(tbl_merge(
  tbls = list(score_prostate, score_colon),
  tab_spanner = c("**Prostate Cancer**", "**Colon Cancer**")
) %>% as_flex_table -> score_pr_col)
```

Warning: fonts used in ‘flextable’ are ignored because the ‘pdflatex’ engine is used and not ‘xelatex’ or ‘lualatex’. You can avoid this warning by using the ‘set_flextable_defaults(fonts_ignore=TRUE)’ command or use a compatible engine by defining ‘latex_engine: xelatex’ in the YAML header of the R Markdown document.

	Prostate Cancer	Colon Cancer
Characteristic	N = 414 ¹	N = 414 ¹
Knowledge		
Poor	160 (39%)	
Good	254 (61%)	
Practice		
Poor	348 (84%)	
Good	66 (16%)	
Knowledge		
Poor		227 (55%)
Good		187 (45%)
Practice		
Poor		323 (78%)
Good		91 (22%)
¹ n (%)		

Age Tables

Breast

```
data %>%
  select(agebrkans, age, Sex, dept, dx_self_cancer, dgraduation) %>%
  tbl_summary(
    by = agebrkans,
    statistic = list(
      all_continuous() ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%)",
    ),
    digits = all_continuous() ~ 2,
    missing = "no",
  ) %>%
  bold_labels() -> age_breast
```

101 missing rows in the "agebrkans" column have been removed.

Cervical

```
data %>%
  select(age_cerv_start, age, Sex, dept, dx_self_cancer, dgraduation ) %>%
  tbl_summary(
    by = age_cerv_start,
    statistic = list(
      all_continuous() ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%)",
    ),
    digits = all_continuous() ~ 2,
    missing = "no",
  ) %>%
  bold_labels() -> age_cervical
```

111 missing rows in the "age_cerv_start" column have been removed.

```
(tbl_merge(
  tbls = list("Breast Cancer" = age_breast, "Cervical Cancer" = age_cervical),
  tab_spanner = c("**Breast Cancer**", "**Cervical Cancer**")
) %>% as_flex_table -> age_br_cer)
```

Warning: fonts used in 'flextable' are ignored because the 'pdflatex' engine is used and not 'xelatex' or 'lualatex'. You can avoid this warning by using the 'set_flextable_defaults(fonts_ignore=TRUE)' command or use a compatible engine by defining 'latex_engine: xelatex' in the YAML header of the R Markdown document.

Characteristic	Breast Cancer			Cervical Cancer	
	<30 N = 176 ¹	30-40 N = 132 ¹	>40 N = 5 ¹	<30 N = 145 ¹	30-40 N = 140 ¹
Age of the patient					
<=40	116 (66%)	88 (67%)	1 (20%)	96 (66%)	90 (64%)
41-50	50 (28%)	27 (20%)	3 (60%)	35 (24%)	41 (29%)
51-60	9 (5.1%)	14 (11%)	1 (20%)	9 (6.2%)	9 (6.4%)
60+	1 (0.6%)	3 (2.3%)	0 (0%)	5 (3.4%)	0 (0%)
Sex of the patient					
Male	107 (61%)	85 (64%)	3 (60%)	79 (54%)	91 (65%)
Female	69 (39%)	47 (36%)	2 (40%)	66 (46%)	49 (35%)
Department					
Non Clinical	20 (15%)	27 (26%)	0 (0%)	26 (23%)	14 (15%)
Clinical	111 (85%)	76 (74%)	5 (100%)	87 (77%)	80 (85%)
Ever been diagnosed any cancer	15 (8.5%)	11 (8.3%)	1 (20%)	12 (8.3%)	8 (5.7%)
Duration of graduation (years)					
<6	33 (24%)	39 (38%)	1 (20%)	34 (30%)	28 (27%)
6-10	54 (39%)	23 (22%)	0 (0%)	44 (39%)	27 (26%)
>10	53 (38%)	42 (40%)	4 (80%)	36 (32%)	47 (46%)

¹n (%)

Prostate

```
data %>%
  select(ageproscs, age, Sex, dept, dx_self_cancer, dgraduation) %>%
  tbl_summary(
    by = ageproscs,
    statistic = list(
      all_continuous() ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = all_continuous() ~ 2,
    missing = "no"
  ) %>%
  bold_labels() -> age_prostate
```

144 missing rows in the "ageproscs" column have been removed.

Colon

```
data %>%
  select(age_colon_ca, age, Sex, dept, dx_self_cancer, dgraduation) %>%
  tbl_summary(
    by = age_colon_ca,
    statistic = list(
      all_continuous() ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = all_continuous() ~ 2,
    missing = "no"
  ) %>%
  bold_labels() -> age_colon
```

127 missing rows in the "age_colon_ca" column have been removed.

```
(tbl_merge(
  tbls = list(age_prostate, age_colon),
  tab_spanner = c("**Prostate Cancer**", "**Colon Cancer**")
) %>% as_flex_table -> age_pr_col)
```

Warning: fonts used in 'flextable' are ignored because the 'pdflatex' engine is used and not 'xelatex' or 'lualatex'. You can avoid this warning by using the 'set_flextable_defaults(fonts_ignore=TRUE)' command or use a compatible engine by defining 'latex_engine: xelatex' in the YAML header of the R Markdown document.

	Prostate Cancer			Colon Cancer	
Characteristic	<30 N = 1 ¹	30-40 N = 86 ¹	>40 N = 183 ¹	<30 N = 17 ¹	30-40 N = 152 ¹
Age of the patient					
<=40	1 (100%)	52 (60%)	116 (63%)	11 (65%)	87 (57%)
41-50	0 (0%)	28 (33%)	44 (24%)	6 (35%)	47 (31%)
51-60	0 (0%)	5 (5.8%)	18 (9.8%)	0 (0%)	15 (9.9%)
60+	0 (0%)	1 (1.2%)	5 (2.7%)	0 (0%)	3 (2.0%)
Sex of the patient					
Male	1 (100%)	58 (67%)	121 (66%)	14 (82%)	107 (70%)
Female	0 (0%)	28 (33%)	62 (34%)	3 (18%)	45 (30%)
Department					
Non Clinical	0 (NA%)	8 (13%)	21 (14%)	2 (17%)	9 (8.3%)
Clinical	0 (NA%)	54 (87%)	124 (86%)	10 (83%)	99 (92%)
Ever been diagnosed any cancer	0 (0%)	8 (9.3%)	15 (8.2%)	0 (0%)	11 (7.2%)
Duration of graduation (years)					
<6	0 (0%)	17 (24%)	38 (27%)	3 (23%)	19 (16%)

Characteristic	Prostate Cancer			Colon Cancer	
	<30 N = 1 ¹	30-40 N = 86 ¹	>40 N = 183 ¹	<30 N = 17 ¹	30-40 N = 152 ¹
6-10	1 (100%)	24 (34%)	45 (32%)	3 (23%)	42 (36%)
>10	0 (0%)	29 (41%)	59 (42%)	7 (54%)	57 (48%)

¹n (%)

Model

Formula

```
fml <- list(
  kb_fml = know_breast ~ age + Sex + Family + dgraduation + dx_self_cancer + know_ca_scrn + att_ca_scrn
  pb_fml = prac_breast ~ age + Sex + Family + dgraduation + dx_self_cancer + know_ca_scrn + att_ca_scrn
  kc_fml = know_colon ~ age + Sex + Family + dgraduation + dx_self_cancer + know_ca_scrn + att_ca_scrn
  pc_fml = prac_colon ~ age + Sex + Family + dgraduation + dx_self_cancer + know_ca_scrn + att_ca_scrn
  kp_fml = know_prostate ~ age + Sex + Family + dgraduation + dx_self_cancer + know_ca_scrn + att_ca_scrn
  pp_fml = prac_prostate ~ age + Sex + Family + dgraduation + dx_self_cancer + know_ca_scrn + att_ca_scrn
  kcr_fml = know_cervical ~ age + Sex + Family + dgraduation + dx_self_cancer + know_ca_scrn + att_ca_scrn
  pcr_fml = prac_cervical ~ age + Sex + Family + dgraduation + dx_self_cancer + know_ca_scrn + att_ca_scrn
)
```

```
models <- lapply(fml, function(formula) {
  glm(formula, data = data_mod, family = binomial)
})
```

```
mod_tables <- lapply(models, function(mod) {
  mod %>%
    tbl_regression(
      exponentiate = T
    ) %>%
    bold_labels() %>%
    modify_table_styling(
      columns = c(estimate),
      rows = reference_row == T,
      missing_symbol = "1"
    )
}) %>% suppressWarnings
```

```
br_cer_kn_mod <- tbl_merge(
  list(mod_tables$kb_fml, mod_tables$kcr_fml),
  tab_spanner = c("**Breast Cancer**", "**Cervical Cancer**")
) %>%
  as_flex_table()

br_cer_pr_mod <- tbl_merge(
  list(mod_tables$pb_fml, mod_tables$pcr_fml),
  tab_spanner = c("**Breast Cancer**", "**Cervical Cancer**")
)
```

```

) %>%
  as_flex_table()

pro_col_kn_mod <- tbl_merge(
  list(mod_tables$kp_fml, mod_tables$kc_fml),
  tab_spanner = c("**Prostate Cancer**", "**Colon Cancer**")
) %>%
  as_flex_table()

pro_col_pr_mod <- tbl_merge(
  list(mod_tables$pp_fml, mod_tables$pc_fml),
  tab_spanner = c("**Prostate Cancer**", "**Colon Cancer**")
) %>%
  as_flex_table()

```

Know-Prac vs Covariate Table

Knowledge

Breast

```

data_mod %>%
  select(know_breast, age, Sex, dept, Family, dx_self_cancer, dgraduation, know_ca_scrn, att_ca_scrn) %>%
  tbl_summary(
    by = know_breast,
    statistic = list(
      all_continuous() ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%"
    ),
    digits = all_continuous() ~ 2,
    missing = "no",
  ) %>%
  bold_labels() -> know_breast_cov

```

Cervical

```

data_mod %>%
  select(know_cervical, age, Sex, dept, Family, dx_self_cancer, dgraduation, know_ca_scrn, att_ca_scrn) %>%
  tbl_summary(
    by = know_cervical,
    statistic = list(
      all_continuous() ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%"
    ),
    digits = all_continuous() ~ 2,
  )

```

```

    missing = "no",
  ) %>%
  bold_labels() -> know_cervical_cov

```

```

(tbl_merge(
  tbls = list("Breast Cancer" = know_breast_cov, "Cervical Cancer" = know_cervical_cov),
  tab_spanner = c("**Breast Cancer**", "**Cervical Cancer**")
) %>% as_flex_table -> know_cov_br_cer)

```

```

## Warning: fonts used in 'flextable' are ignored because the 'pdflatex' engine is
## used and not 'xelatex' or 'lualatex'. You can avoid this warning by using the
## 'set_flextable_defaults(fonts_ignore=TRUE)' command or use a compatible engine
## by defining 'latex_engine: xelatex' in the YAML header of the R Markdown
## document.

```

Characteristic	Breast Cancer		Cervical Cancer	
	Poor N = 114 ¹	Good N = 300 ¹	Poor N = 120 ¹	Good N = 294 ¹
Age of the patient				
<=40	72 (63%)	196 (65%)	75 (63%)	193 (66%)
41-50	28 (25%)	76 (25%)	27 (23%)	77 (26%)
51-60	11 (9.6%)	24 (8.0%)	16 (13%)	19 (6.5%)
60+	3 (2.6%)	4 (1.3%)	2 (1.7%)	5 (1.7%)
Sex of the patient				
Male	78 (68%)	186 (62%)	86 (72%)	178 (61%)
Female	36 (32%)	114 (38%)	34 (28%)	116 (39%)
Department				
Non Clinical	13 (11%)	53 (18%)	17 (14%)	49 (17%)
Clinical	101 (89%)	247 (82%)	103 (86%)	245 (83%)
Family H/O cancer	22 (19%)	46 (15%)	26 (22%)	42 (14%)
Ever been diagnosed any cancer	6 (5.3%)	27 (9.0%)	11 (9.2%)	22 (7.5%)
Duration of graduation (years)				
<6	27 (24%)	95 (32%)	30 (25%)	92 (31%)
6-10	39 (34%)	85 (28%)	36 (30%)	88 (30%)
>10	48 (42%)	120 (40%)	54 (45%)	114 (39%)
Know about cancer screening	83 (73%)	264 (88%)	99 (83%)	248 (84%)
Think cancer screening will usefull	88 (77%)	286 (95%)	100 (83%)	274 (93%)

¹n (%)

```
rm(know_breast_cov, know_cervical_cov)
```

Prostate

```
data_mod %>%
  select(know_prostate, age, Sex, dept, Family, dx_self_cancer, dgraduation, know_ca_scrn, att_ca_scrn) %>%
  tbl_summary(
    by = know_prostate,
    statistic = list(
      all_continuous() ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%"
    ),
    digits = all_continuous() ~ 2,
    missing = "no",
  ) %>%
  bold_labels() -> know_prostate_cov
```

Colon

```
data_mod %>%
  select(know_colon, age, Sex, dept, Family, dx_self_cancer, dgraduation, know_ca_scrn, att_ca_scrn) %>%
  tbl_summary(
    by = know_colon,
    statistic = list(
      all_continuous() ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%"
    ),
    digits = all_continuous() ~ 2,
    missing = "no",
  ) %>%
  bold_labels() -> know_colon_cov
```

```
(tbl_merge(
  tbls = list(know_prostate_cov, know_colon_cov),
  tab_spanner = c("**Prostate Cancer**", "**Colon Cancer**")
) %>% as_flex_table -> know_cov_pr_col)
```

```
## Warning: fonts used in 'flextable' are ignored because the 'pdflatex' engine is
## used and not 'xelatex' or 'lualatex'. You can avoid this warning by using the
## 'set_flextable_defaults(fonts_ignore=TRUE)' command or use a compatible engine
## by defining 'latex_engine: xelatex' in the YAML header of the R Markdown
## document.
```


Characteristic	Prostate Cancer		Colon Cancer	
	Poor N = 160 ¹	Good N = 254 ¹	Poor N = 227 ¹	Good N = 187 ¹
Age of the patient				
<=40	110 (69%)	158 (62%)	141 (62%)	127 (68%)
41-50	34 (21%)	70 (28%)	60 (26%)	44 (24%)
51-60	14 (8.8%)	21 (8.3%)	21 (9.3%)	14 (7.5%)
60+	2 (1.3%)	5 (2.0%)	5 (2.2%)	2 (1.1%)
Sex of the patient				
Male	95 (59%)	169 (67%)	140 (62%)	124 (66%)
Female	65 (41%)	85 (33%)	87 (38%)	63 (34%)
Department				
Non Clinical	30 (19%)	36 (14%)	41 (18%)	25 (13%)
Clinical	130 (81%)	218 (86%)	186 (82%)	162 (87%)
Family H/O cancer	33 (21%)	35 (14%)	46 (20%)	22 (12%)
Ever been diagnosed any cancer	11 (6.9%)	22 (8.7%)	21 (9.3%)	12 (6.4%)
Duration of graduation (years)				
<6	53 (33%)	69 (27%)	63 (28%)	59 (32%)
6-10	46 (29%)	78 (31%)	66 (29%)	58 (31%)
>10	61 (38%)	107 (42%)	98 (43%)	70 (37%)
Know about cancer screening	134 (84%)	213 (84%)	185 (81%)	162 (87%)
Think cancer screening will usefull	139 (87%)	235 (93%)	199 (88%)	175 (94%)

¹n (%)

```
rm(know_prostate_cov, know_colon_cov)
```

Practice

Breast

```
data_mod %>%
  select(prac_breast, age, Sex, dept, Family, dx_self_cancer, dgraduation, know_ca_scrn, att_ca_scrn) %>%
  tbl_summary(
    by = prac_breast,
    statistic = list(
      all_continuous() ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%)",
    ),
    digits = all_continuous() ~ 2,
    missing = "no",
```

```
) %>%
bold_labels() -> prac_breast_cov
```

Cervical

```
data_mod %>%
  select(prac_cervical,age,Sex,dept,Family,dx_self_cancer,dgraduation,know_ca_scrn,att_ca_scrn) %>%
  tbl_summary(
    by = prac_cervical,
    statistic = list(
      all_continuous() ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = all_continuous() ~ 2,
    missing = "no",
  ) %>%
  bold_labels() -> prac_cervical_cov
```

```
(tbl_merge(
  tbls = list(prac_breast_cov, prac_cervical_cov),
  tab_spanner = c("**Breast Cancer**", "**Cervical Cancer**")
) %>% as_flex_table -> prac_cov_br_cer)
```

```
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```

Characteristic	Breast Cancer		Cervical Cancer	
	Poor N = 353 ¹	Good N = 61 ¹	Poor N = 347 ¹	Good N = 67 ¹
Age of the patient				
<=40	229 (65%)	39 (64%)	228 (66%)	40 (60%)
41-50	89 (25%)	15 (25%)	86 (25%)	18 (27%)
51-60	29 (8.2%)	6 (9.8%)	27 (7.8%)	8 (12%)
60+	6 (1.7%)	1 (1.6%)	6 (1.7%)	1 (1.5%)
Sex of the patient				
Male	229 (65%)	35 (57%)	220 (63%)	44 (66%)
Female	124 (35%)	26 (43%)	127 (37%)	23 (34%)
Department				
Non Clinical	57 (16%)	9 (15%)	63 (18%)	3 (4.5%)
Clinical	296 (84%)	52 (85%)	284 (82%)	64 (96%)

Characteristic	Breast Cancer		Cervical Cancer	
	Poor N = 353 ¹	Good N = 61 ¹	Poor N = 347 ¹	Good N = 67 ¹
Family H/O cancer	51 (14%)	17 (28%)	56 (16%)	12 (18%)
Ever been diagnosed any cancer	23 (6.5%)	10 (16%)	28 (8.1%)	5 (7.5%)
Duration of graduation (years)				
<6	100 (28%)	22 (36%)	114 (33%)	8 (12%)
6-10	108 (31%)	16 (26%)	95 (27%)	29 (43%)
>10	145 (41%)	23 (38%)	138 (40%)	30 (45%)
Know about cancer screening	296 (84%)	51 (84%)	292 (84%)	55 (82%)
Think cancer screening will usefull	317 (90%)	57 (93%)	316 (91%)	58 (87%)

¹n (%)

```
rm(prac_breast_cov, prac_cervical_cov)
```

Prostate

```
data_mod %>%
  select(prac_prostate,age,Sex,dept,Family,dx_self_cancer,dgraduation,know_ca_scrn,att_ca_scrn) %>%
  tbl_summary(
    by = prac_prostate,
    statistic = list(
      all_continuous() ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = all_continuous() ~ 2,
    missing = "no",
  ) %>%
  bold_labels() -> prac_prostate_cov
```

Colon

```
data_mod %>%
  select(prac_colon,age,Sex,dept,Family,dx_self_cancer,dgraduation,know_ca_scrn,att_ca_scrn) %>%
  tbl_summary(
    by = prac_colon,
    statistic = list(
      all_continuous() ~ "{mean} ± {sd}",
      all_categorical() ~ "{n} ({p}%)"
    ),
    digits = all_continuous() ~ 2,
    missing = "no",
  ) %>%
  bold_labels() -> prac_colon_cov
```

```
(tbl_merge(
  tbls = list(prac_prostate_cov, prac_colon_cov),
  tab_spanner = c("**Prostate Cancer**", "**Colon Cancer**")
) %>% as_flex_table -> prac_cov_pr_col)
```

```
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```

	Prostate Cancer		Colon Cancer	
Characteristic	Poor N = 348 ¹	Good N = 66 ¹	Poor N = 323 ¹	Good N = 91 ¹
Age of the patient				
<=40	229 (66%)	39 (59%)	209 (65%)	59 (65%)
41-50	85 (24%)	19 (29%)	80 (25%)	24 (26%)
51-60	28 (8.0%)	7 (11%)	27 (8.4%)	8 (8.8%)
60+	6 (1.7%)	1 (1.5%)	7 (2.2%)	0 (0%)
Sex of the patient				
Male	223 (64%)	41 (62%)	212 (66%)	52 (57%)
Female	125 (36%)	25 (38%)	111 (34%)	39 (43%)
Department				
Non Clinical	57 (16%)	9 (14%)	55 (17%)	11 (12%)
Clinical	291 (84%)	57 (86%)	268 (83%)	80 (88%)
Family H/O cancer	60 (17%)	8 (12%)	54 (17%)	14 (15%)
Ever been diagnosed any cancer	24 (6.9%)	9 (14%)	23 (7.1%)	10 (11%)
Duration of graduation (years)				
<6	108 (31%)	14 (21%)	107 (33%)	15 (16%)
6-10	102 (29%)	22 (33%)	86 (27%)	38 (42%)
>10	138 (40%)	30 (45%)	130 (40%)	38 (42%)
Know about cancer screening	284 (82%)	63 (95%)	265 (82%)	82 (90%)
Think cancer screening will usefull	312 (90%)	62 (94%)	287 (89%)	87 (96%)

¹n (%)

```
rm(prac_prostate_cov, prac_colon_cov)
```

Breast - Cervical Tables

```
read_docx() %>%
  body_add_par(value = "Demographic") %>%
  body_add_flextable(value = demo_tab) %>%
  body_add_par(value = "") %>%
  body_add_par(value = "Knowledge Table") %>%
  body_add_flextable(value = know_tab) %>%
  body_add_par(value = "") %>%
  body_add_par(value = "Score Table") %>%
  body_add_par(value = "Knowledge") %>%
  body_add_flextable(value = know_cov_br_cer) %>%
  body_add_par(value = "") %>%
  body_add_par(value = "Practice") %>%
  body_add_flextable(value = prac_cov_br_cer) %>%
  body_add_par(value = "") %>%
  body_add_par(value = "Age of start of cancer screening") %>%
  body_add_flextable(value = age_br_cer) %>%
  body_add_par(value = "") %>%
  body_add_par(value = "Knowledge Model") %>%
  body_add_flextable(value = br_cer_kn_mod) %>%
  body_add_par(value = "") %>%
  body_add_par(value = "Practice Model") %>%
  body_add_flextable(value = br_cer_pr_mod) %>%
  body_add_par(value = "") %>%
  print(target = "doc/Breast_Cervical_Tables.docx")
```

Prostate - Colon Tables

```
read_docx() %>%
  body_add_par(value = "Demographic") %>%
  body_add_flextable(value = demo_tab) %>%
  body_add_par(value = "") %>%
  body_add_par(value = "Knowledge Table") %>%
  body_add_flextable(value = know_tab) %>%
  body_add_par(value = "") %>%
  body_add_par(value = "Score Table") %>%
  body_add_par(value = "Knowledge") %>%
  body_add_flextable(value = know_cov_pr_col) %>%
  body_add_par(value = "") %>%
  body_add_par(value = "Practice") %>%
  body_add_flextable(value = prac_cov_pr_col) %>%
  body_add_par(value = "") %>%
  body_add_par(value = "Age of start of cancer screening") %>%
  body_add_flextable(value = age_pr_col) %>%
  body_add_par(value = "") %>%
  body_add_par(value = "Knowledge Model") %>%
  body_add_flextable(value = pro_col_kn_mod) %>%
  body_add_par(value = "") %>%
  body_add_par(value = "Practice Model") %>%
  print(target = "doc/Prostate_Colon_Tables.docx")
```

```
body_add_flextable(value = pro_col_pr_mod) %>%  
body_add_par(value = "") %>%  
print(target = "doc/Prostate_Colon_Tables.docx")
```